Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 02-03-2026

Responsive E-Commerce Interface with State-Persistent Cart

SDG: SDG No – SDG Title(SDG 8 – Decent Work and Economic Growth)

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INTRODUCTION

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- In today's digital era, traditional shopping methods lack convenience, personalization, and real-time accessibility.
- Customers expect fast, secure, and user-friendly online platforms for purchasing products.
- This project addresses the need for a structured and efficient e-commerce system.
- The system allows users to register, log in, browse products, add items to cart, and place orders.
- It ensures secure authentication using session management and password encryption.
- The application integrates PHP and MySQL to manage users, products, cart, and orders efficiently.
- This project demonstrates real-world implementation of web technologies used in commercial websites.
- Overall, it provides a scalable foundation for building modern online shopping platforms.

PROBLEM STATEMENT

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- Many small businesses still rely on traditional or semi-manual methods to manage product sales and customer orders.
- This leads to inefficiencies in handling inventory, tracking customer data, and processing transactions.
- Manual systems increase the risk of errors, data loss, and delayed order management.
- There is a growing need for an automated and centralized online shopping system.
- Customers expect secure login, easy product browsing, and smooth checkout processes.
- Businesses require a reliable database system to manage users, products, and orders efficiently.
- Hence, a web-based E-Commerce Management System is proposed to solve these challenges.
- The system aims to improve accuracy, accessibility, security, and overall operational efficiency.

OBJECTIVE(S)

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- ✓ To design and develop a secure web-based E-Commerce Management System using PHP and MySQL for efficient handling of users, products, cart, and orders.
- ✓ To implement user authentication and session management to ensure secure login, registration, and data protection.
- ✓ To provide an interactive and user-friendly interface that allows customers to browse products, add items to cart, and complete purchases smoothly.
- ✓ To automate order processing and maintain accurate transaction records using a structured database system.
- ✓ To reduce manual work and improve operational efficiency through a centralized online platform.

PROPOSED SYSTEM

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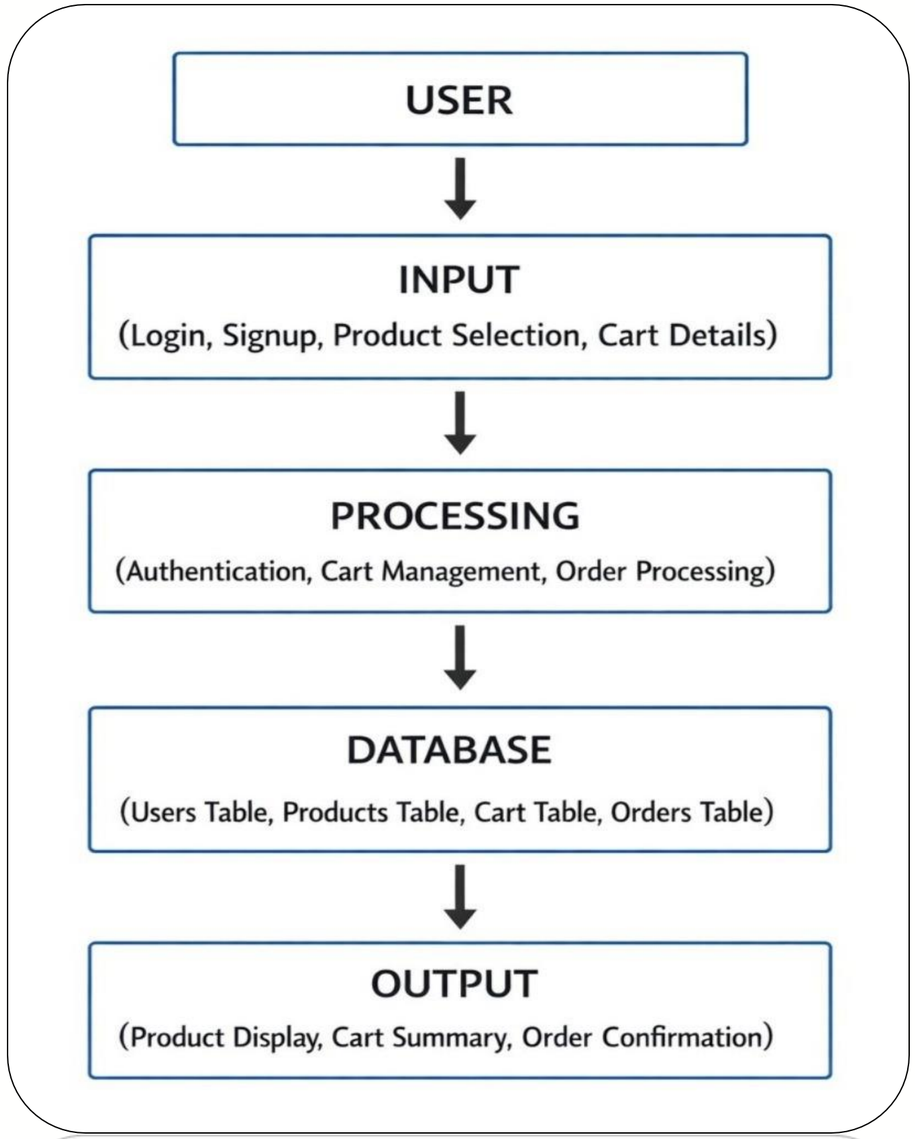
- The proposed solution is a web-based E-Commerce Management System developed using PHP and MySQL.
- The system provides secure user registration and login with session management and password encryption.
- It allows customers to browse products, add items to the cart, and place orders through an automated checkout process.
- All user, product, cart, and order data are stored and managed efficiently using a structured database.
- This system solves issues of manual record-keeping, data redundancy, and transaction errors found in traditional systems.
- It reduces paperwork and improves speed, accuracy, and reliability in managing sales and customer information.
- Compared to existing manual or semi-automated systems, it offers better data security, centralized access, and real-time updates.
- Overall, the solution provides a scalable, efficient, and user-friendly platform suitable for modern online businesses.

PROPOSED SYSTEM ARCHITECTURE



Input:

Users enter registration details, login credentials, select products, add items to the cart, and proceed to checkout through the website interface.

Processing:

The system verifies user authentication, manages cart operations, calculates total amounts, and processes orders using PHP and session management.

Database:

All data such as users, products, cart items, and orders are stored and managed securely in the MySQL database.

Output:

The system displays product listings, cart details, order confirmations, and user information dynamically on the web pages.

WORKFLOW

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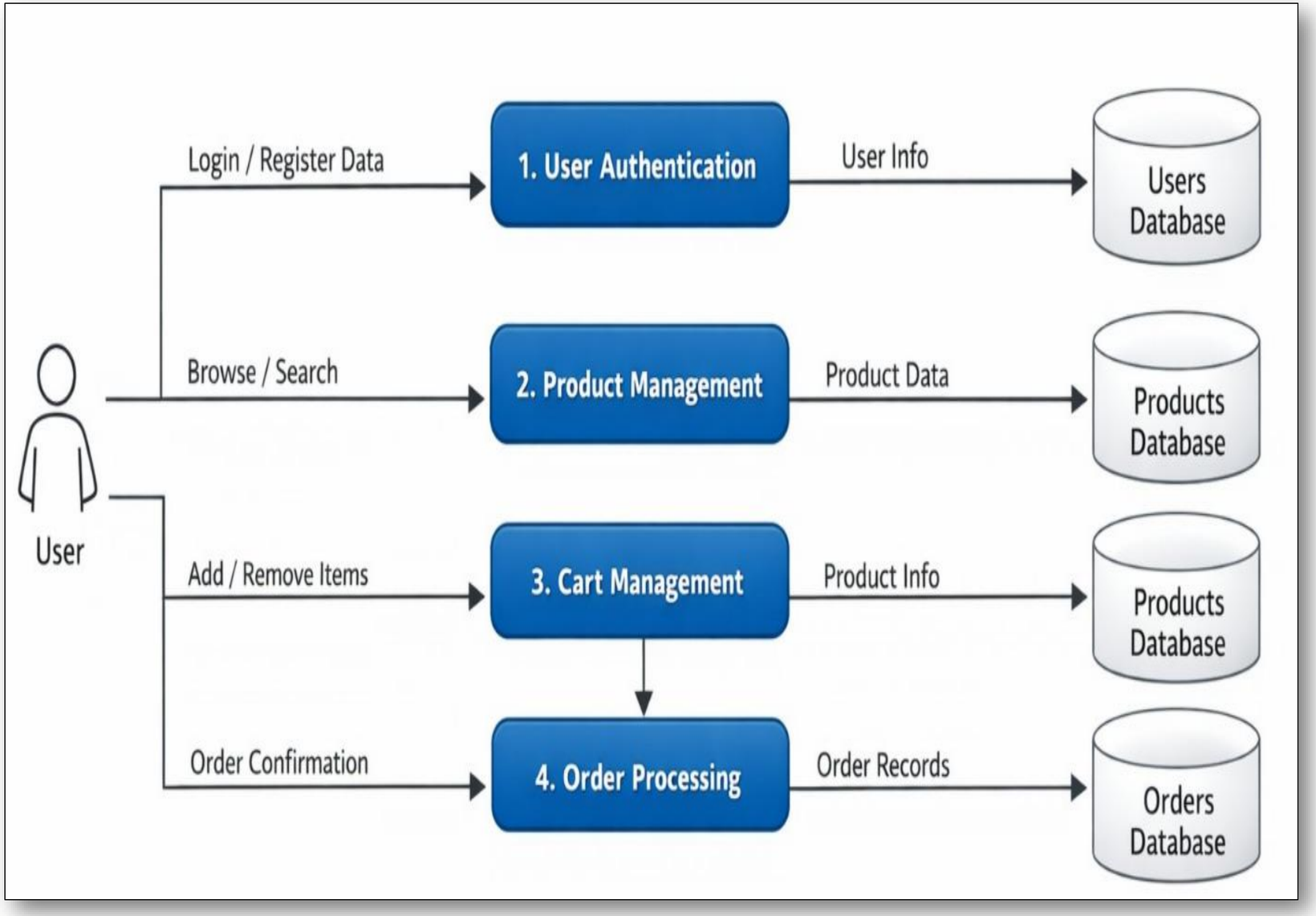
- The User logs in and searches for products.
- The system retrieves product details from the MySQL Database (via XAMPP server).
- The user adds items to cart and places an order.
- The system processes the request and stores order details in the database.
- The Admin manages products and monitors orders.
- The system generates output such as order confirmation and updated records.



TECHNOLOGIES USED

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💻 Frontend:

🌐 **HTML5** – Used to structure the web pages such as login, homepage, cart, and orders.
🎨 **CSS3** – Used to design and style the website with a professional and responsive layout.
⚡ **JavaScript** – Used for basic interactivity and improving user experience.

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🗄️ Backend:

🐘 **PHP** – Used to handle server-side logic including authentication, session management, cart operations, and order processing.

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🗄️ Database:

🛢️ **MySQL** – Used to store and manage user details, product information, cart data, and order records securely.

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🛠 Tools:

🔧 **XAMPP** – Used as a local server environment integrating Apache and MySQL to run the project.
💻 **Visual Studio Code** – Used as the code editor for development.
🌍 **Web Browser (Chrome/Edge)** – Used for testing and running the application.

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📜 Software Requirements:

- Windows 10/11 Operating System
- 🔧 XAMPP Server
- 🌐 Web Browser
- 📝 Code Editor (Visual Studio Code)

MODULE DESCRIPTION

Module 1: User Management

- Handles user registration and login with secure authentication using password hashing and session management.
- It ensures that only authorized users can access the system.

Module 2: Product Management

- Displays available products with images, descriptions, and prices.
- Allows users to browse and select products easily.

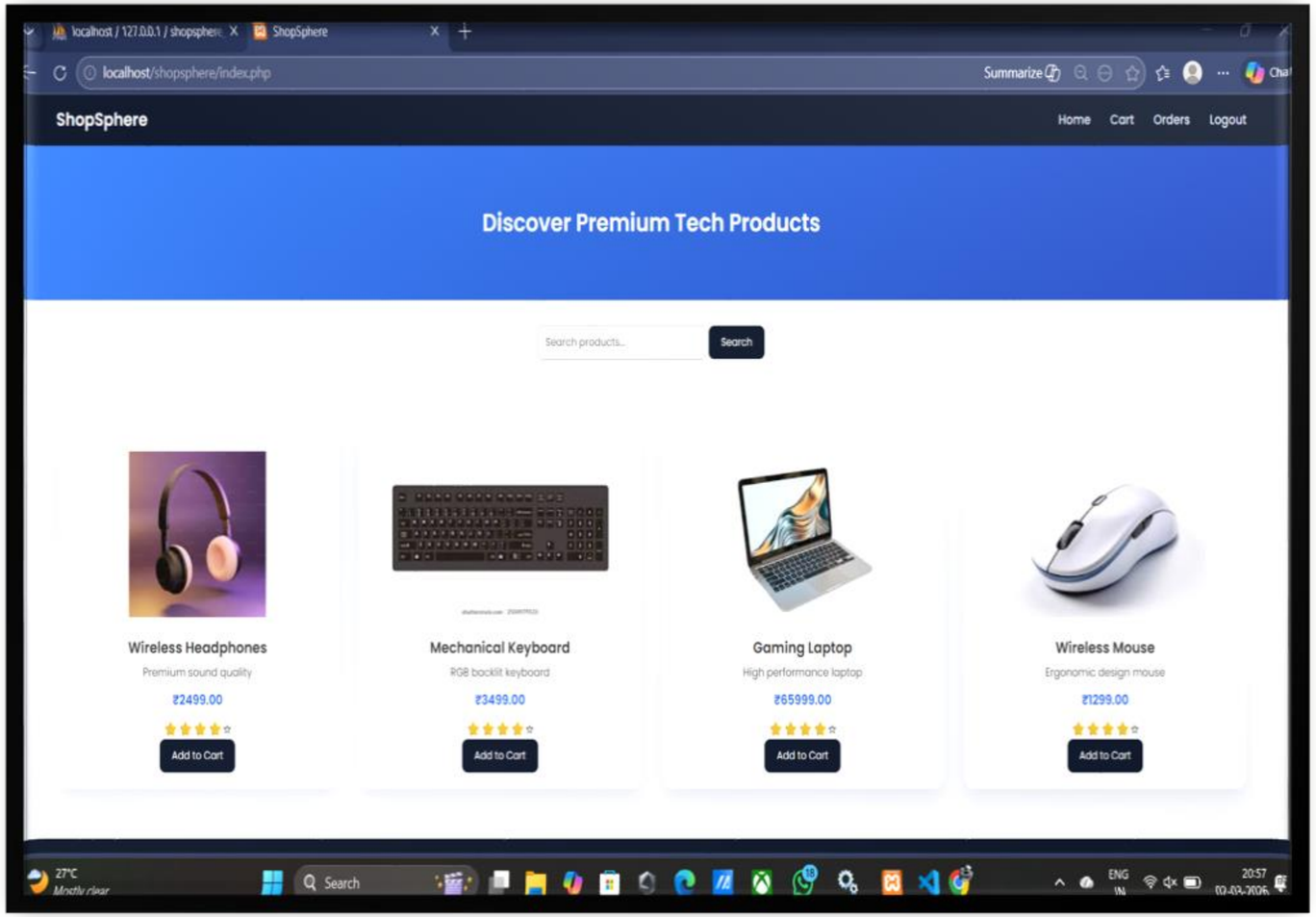
Module 3: Cart & Order Processing

- Manages adding products to cart, calculating total price, processing checkout, and storing order details in the database.

Module 4: Database Management

- Handles storage and retrieval of user, product, cart, and order data using MySQL, ensuring data integrity and efficient transaction management.

IMPLEMENTATION SCREENSHOTS



Home Page – Product Dashboard

- The Home Page acts as the main interface of the system where users can view all available products.
- Each product displays its name, description, price, rating, and an "Add to Cart" option.
- The product data is dynamically fetched from the MySQL database using PHP.
- The search feature allows users to filter products based on keywords.

IMPLEMENTATION SCREENSHOTS

LOGIN & SIGNUP - User Authentication Module

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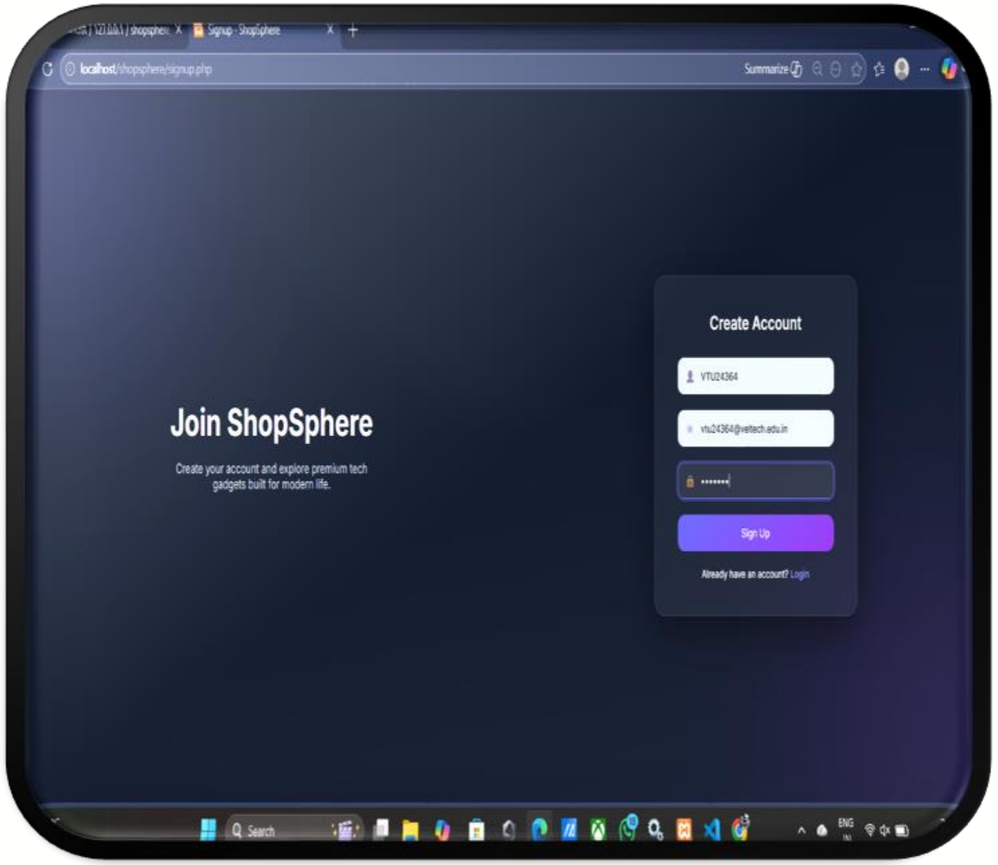
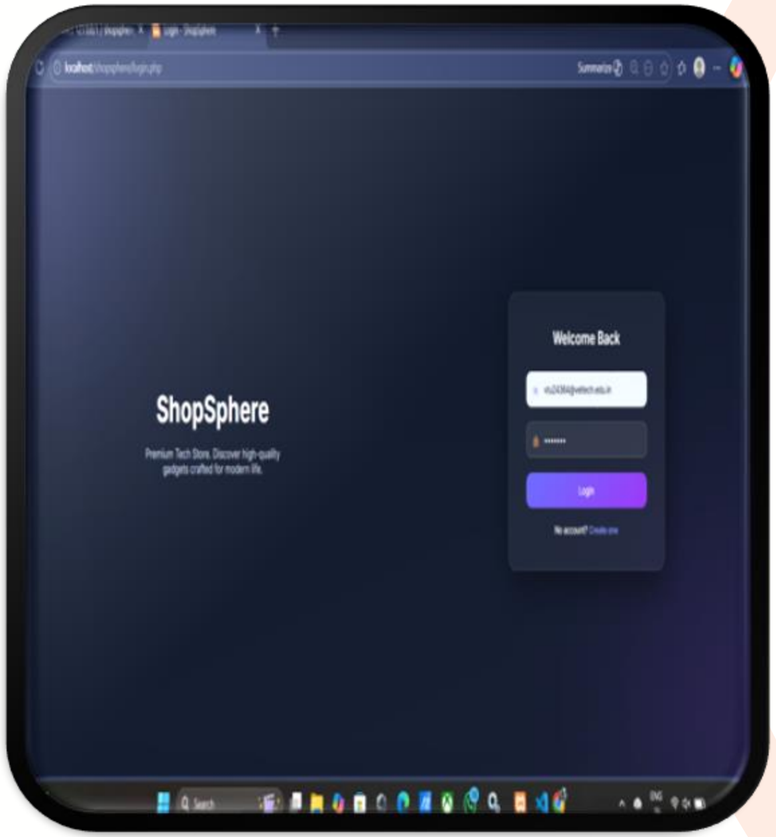


- ✓ During signup, user details like username, email, and password are stored in the database.
- ✓ Passwords are encrypted using PHP password hashing for security.



- ✓ During login, the entered credentials are verified with database records before granting access.

IMPLEMENTATION SCREENSHOTS

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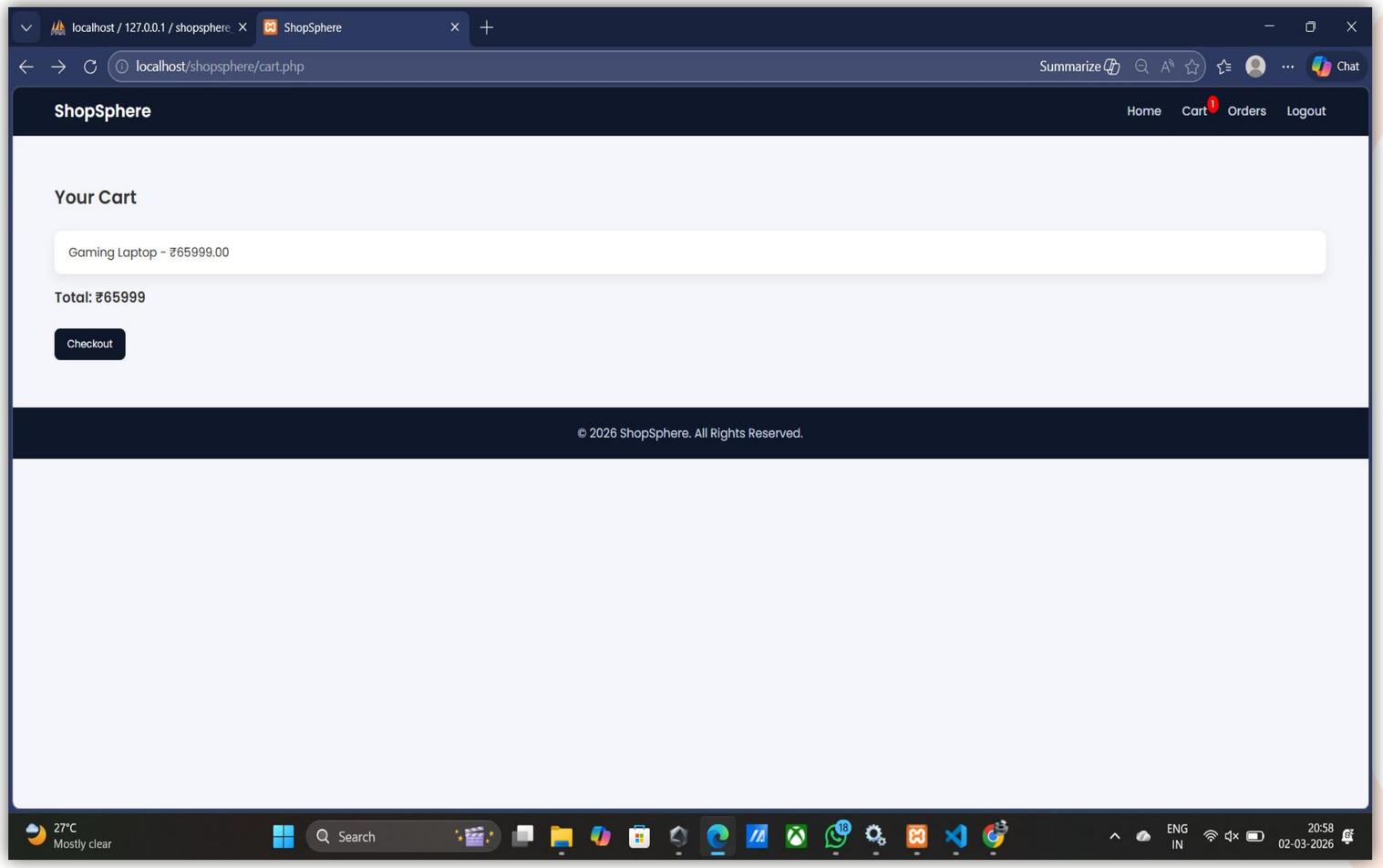
🛒 SLIDE 3 – CART & CHECKOUT

Cart & Checkout Module

- ❖ The Cart Module allows users to add selected products for purchase.
- ❖ When a product is added to the cart, the system stores it in the cart table linked to the user ID.
- ❖ The total amount is calculated dynamically.
- ❖ During checkout, order details are stored in the orders table along with order status and timestamp.



IMPLEMENTATION SCREENSHOTS

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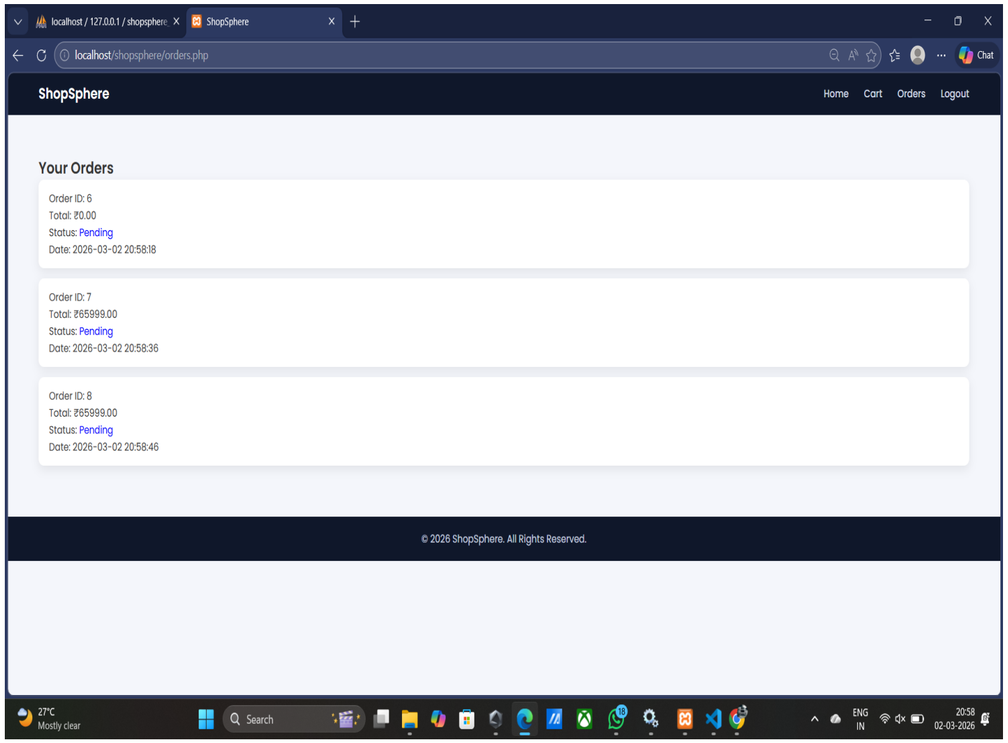

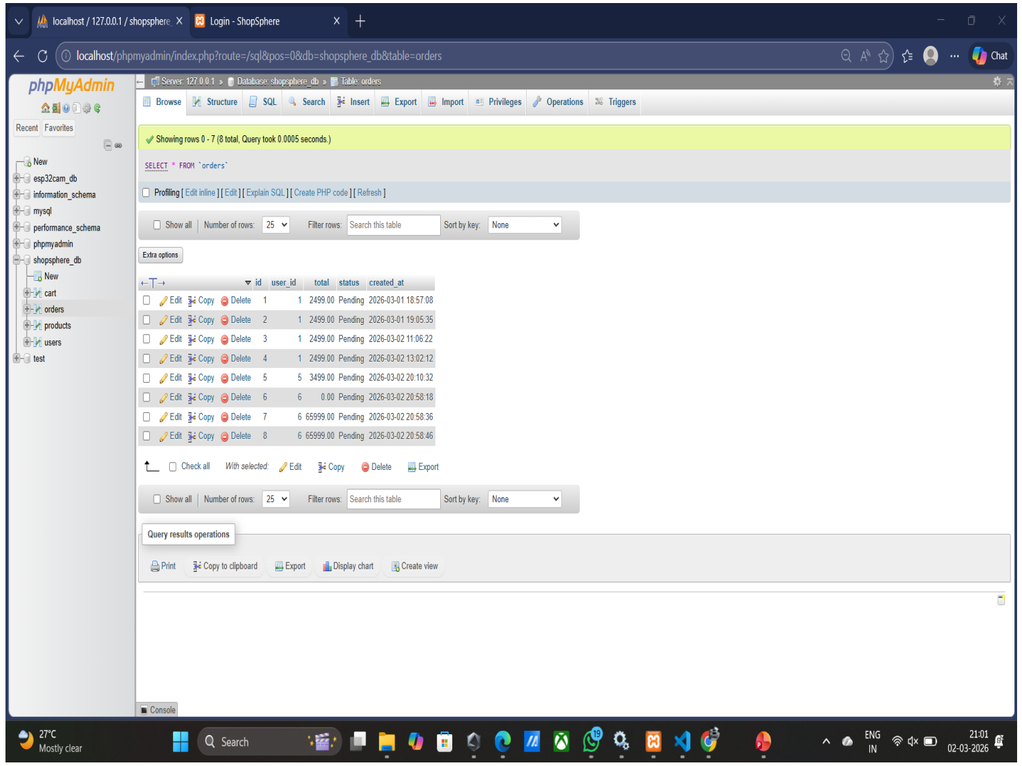

ORDERS & DATABASE - Order Output & Database Integration

- The Orders Page displays all previous transactions of the logged-in user.
- It shows order ID, total amount, order status (Pending), and date of purchase.
- This data is retrieved from the MySQL database using SQL queries.
- The backend uses PHP to connect and interact with the database for storing and retrieving records.

ADVANTAGES & LIMITATIONS

✅ Advantages

- ✓ Provides secure user authentication using password encryption and session management.
- ✓ Reduces manual work by automating product browsing, cart handling, and order processing.
- ✓ Ensures accurate data storage using MySQL database integration.
- ✓ Offers a user-friendly and responsive interface for better customer experience.
- ✓ Centralized system for managing users, products, and orders efficiently.

⚠ Limitations

- ❖ Currently designed for small-scale business and local server deployment (XAMPP).
- ❖ Does not include online payment gateway integration.
- ❖ Limited admin control features in the basic version.
- ❖ Requires internet connectivity for real-time deployment.

Thank you